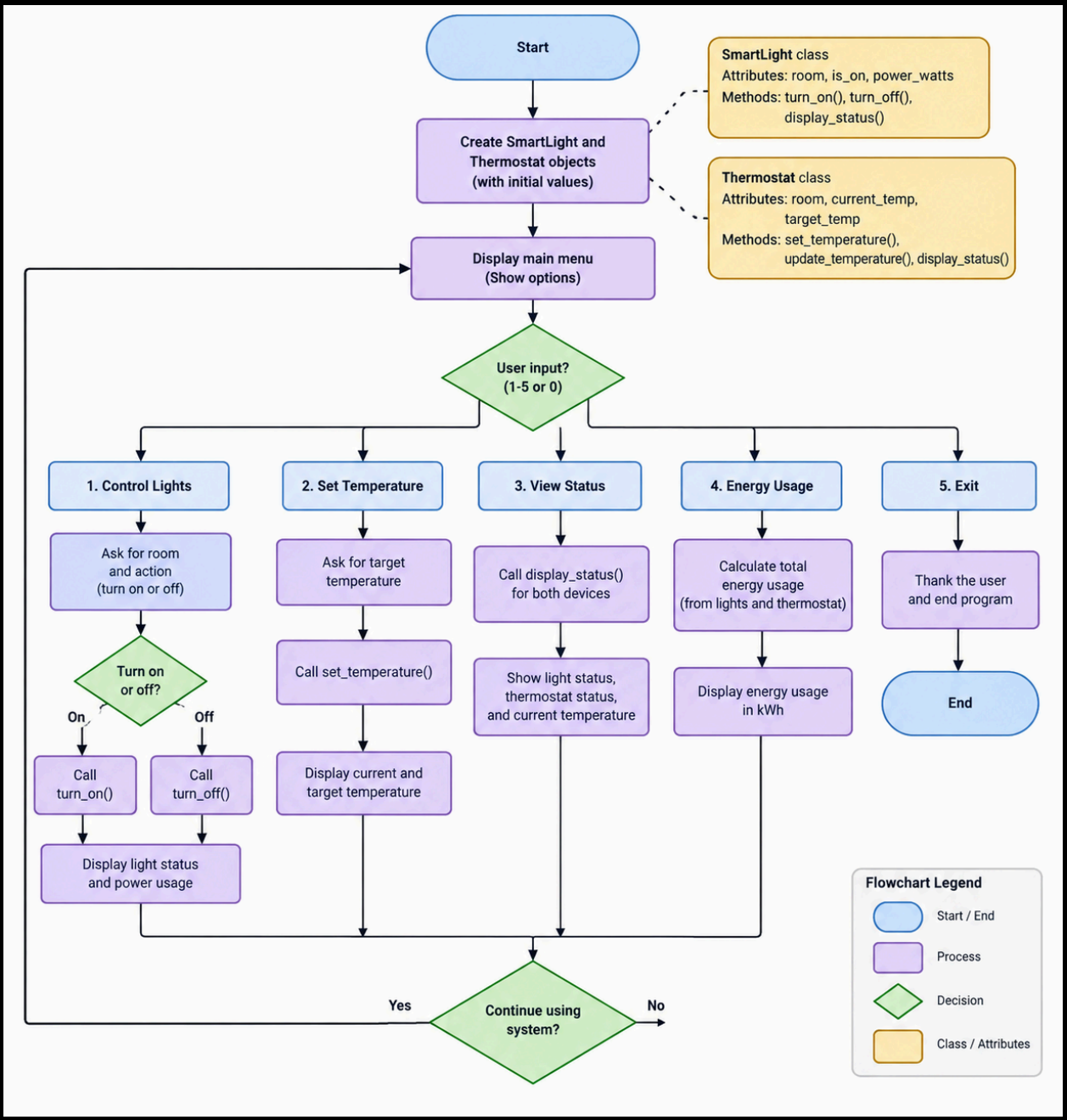

Start
SmartLight class
Attributes: room, is_on, power_watts
Methods: turn_on(), turn_off(), display_status()
Create SmartLight and Thermostat objects (with initial values)
Thermostat class
Attributes: room, current_temp, target_temp
Methods: set_temperature(), update_temperature(), display_status()
Display main menu (Show options)
User input? (1-5 or 0)
1. Control Lights
2. Set Temperature
3. View Status
4. Energy Usage
5. Exit
Ask for room and action (turn on or off)
Ask for target temperature
Call display_status() for both devices
Calculate total energy usage (from lights and thermostat)
Thank the user and end program
Turn on or off?
On
Off
Call turn_on()
Call turn_off()
Call set_temperature()
Display current and target temperature
Show light status, thermostat status, and current temperature
Display energy usage in kWh
End
Display light status and power usage
Yes
Continue using system?
No
Flowchart Legend
Start / End
Process
Decision
Class / Attributes